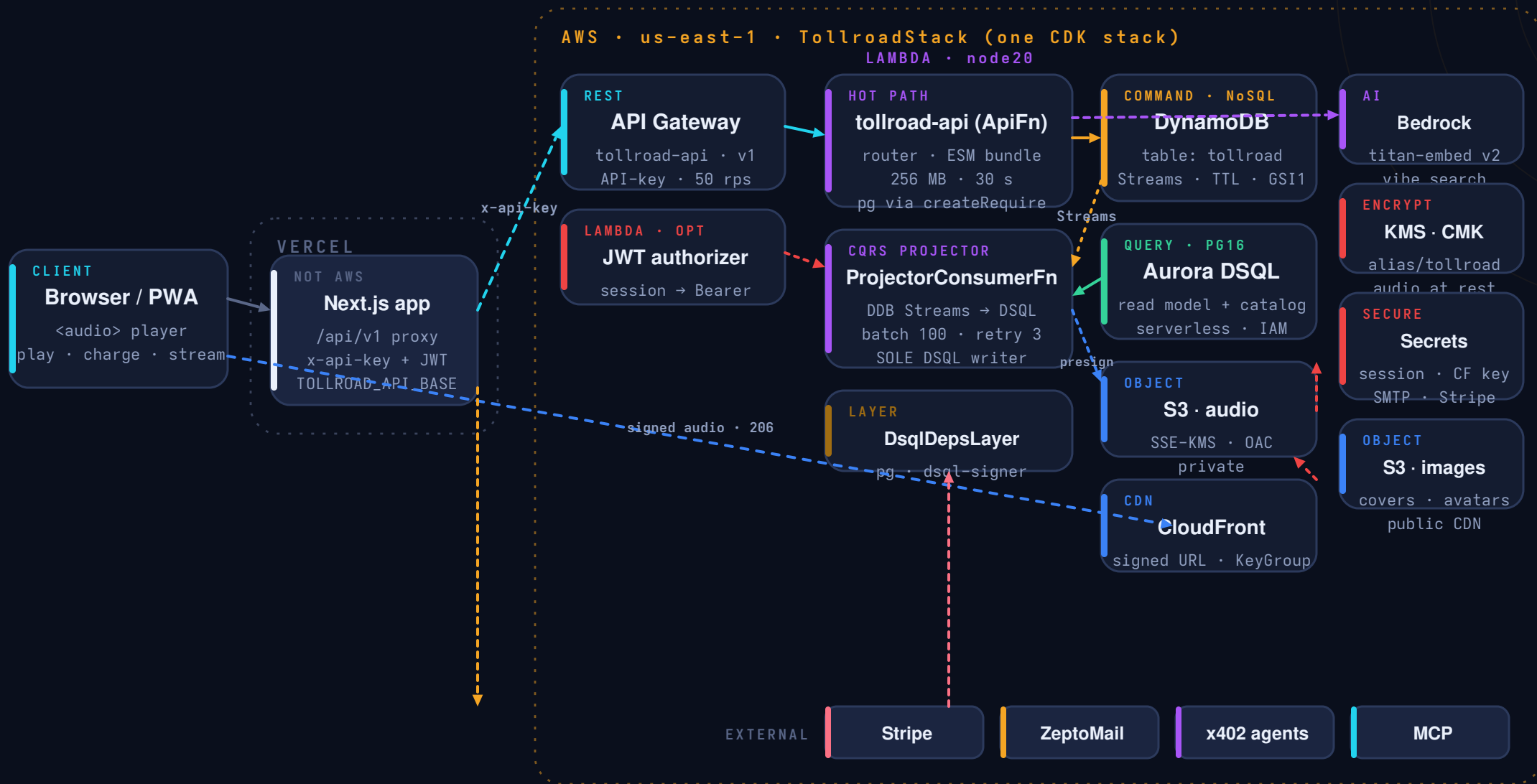
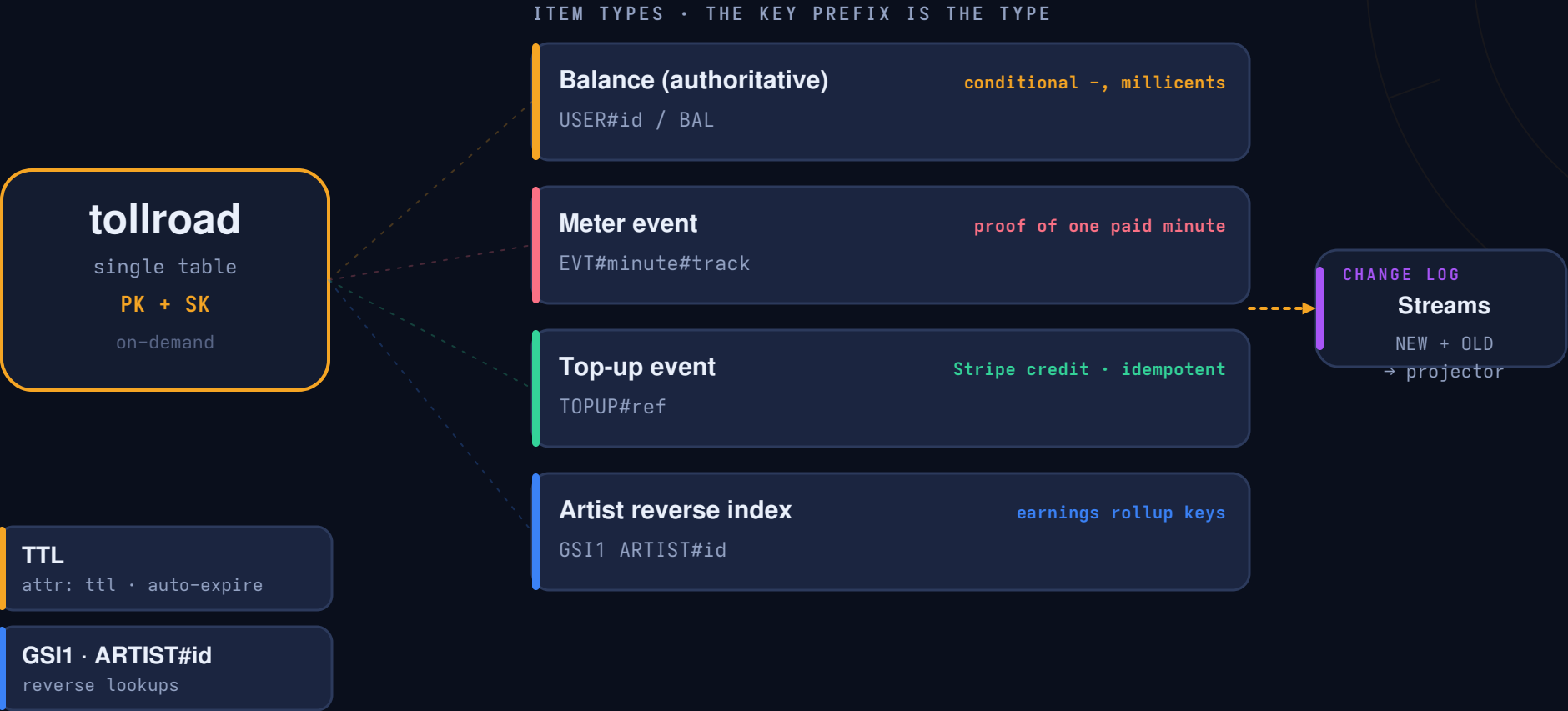


The entire AWS system



Three planes: a synchronous request/billing path (Browser → Vercel proxy → API Gateway → tollroad-api λ), an asynchronous CQRS projection (DynamoDB Streams → ProjectorConsumerFn → Aurora DSQL), and an audio plane (KMS → S3 → signed CloudFront). One region, one CDK stack.

The command side — money, at write speed



THE ATOMIC WRITE · one TransactWriteItems

1 · conditional debit USER#id/BAL -1 metered minute (stop-at-zero)

2 · append EVT#minute#track METER event

balance & proof-of-pay move together, or not at all — never a partial charge

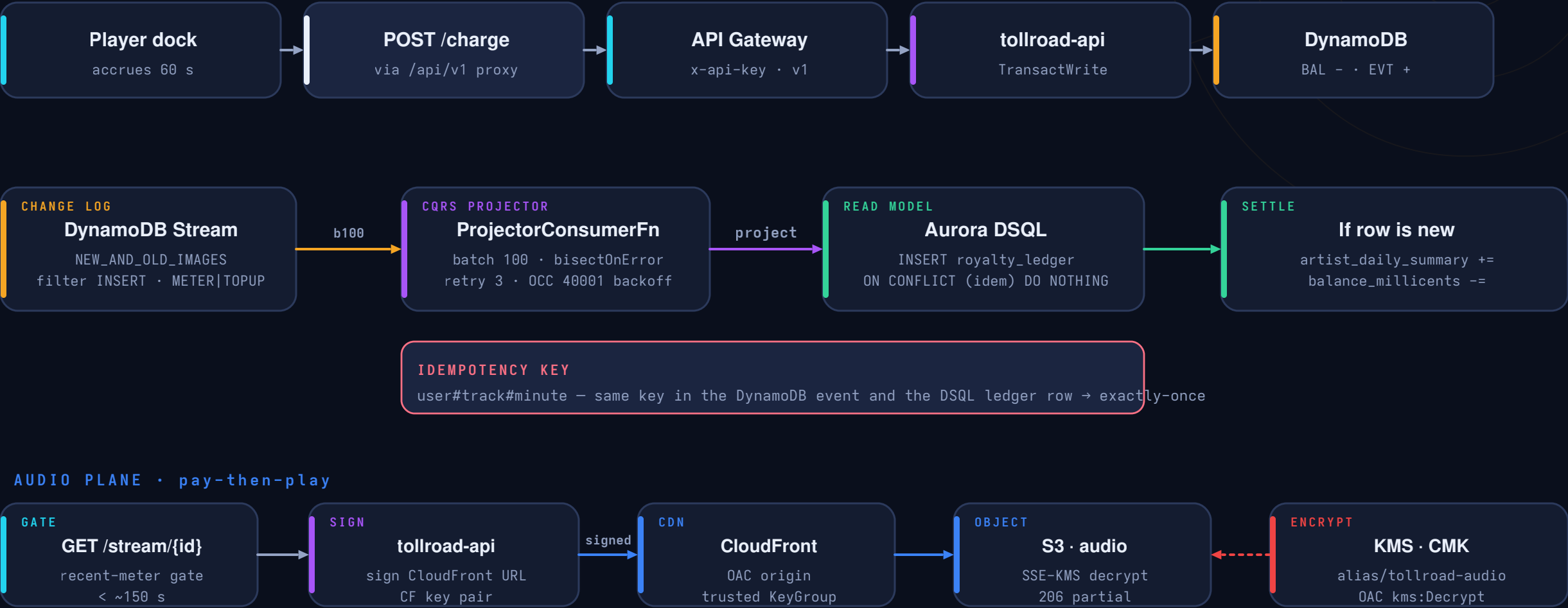
DynamoDB is the authoritative ledger of record for balance: every play is an atomic, conditional decrement that stops exactly at zero. A single table holds balance, meter events and top-ups; the key prefix is the type, TTL ages events out, and the Stream is the change-log that feeds Aurora DSQL.

The query side — relational truth



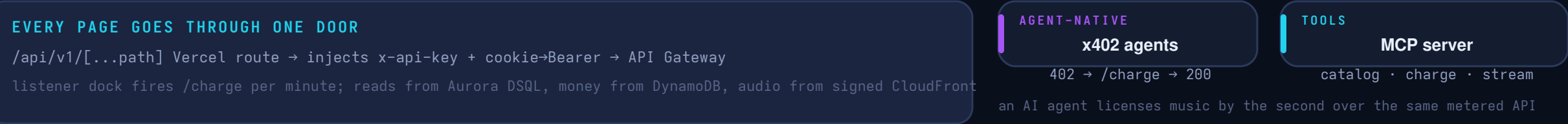
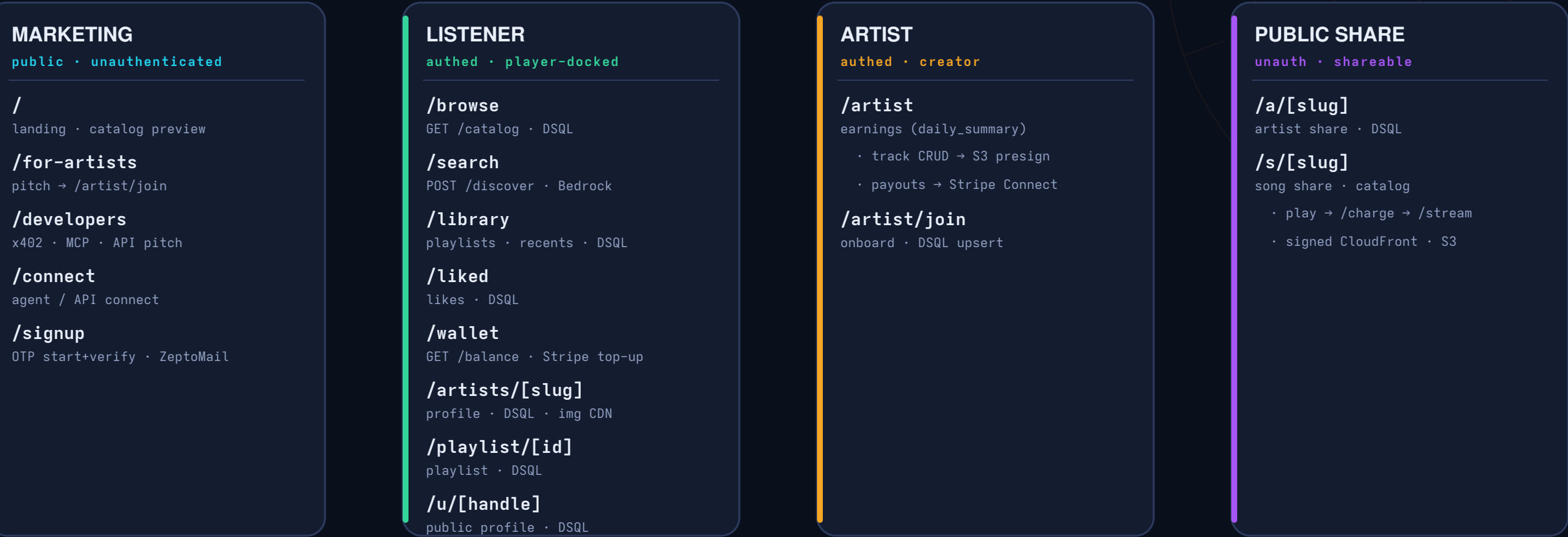
Everything you read — the catalog, an artist's earnings, the royalty ledger, top-ups, the reconciled balance — lives in Aurora DSQL. The API connects as admin to read; only the projector writes, under a least-privilege role with a fresh IAM token per connection. Serverless: it scales to zero, so idle costs nothing.

How a play becomes money



The player meters real playback and fires POST /charge every whole minute — one atomic DynamoDB write debits the balance and records proof of payment together. The Stream replays each event into Aurora DSQL idempotently: the royalty ledger, the artist's daily earnings, the reconciled balance. Audio is pay-then-play: a signed CloudFront URL gated on a recent charge.

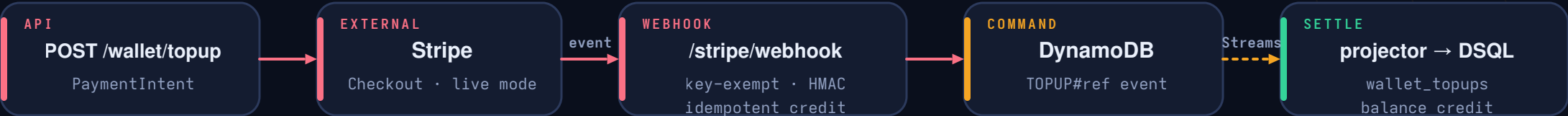
Every page, and the AWS it touches



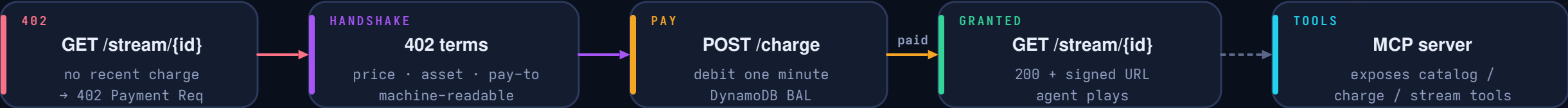
Four surfaces — marketing, the docked listener app, the artist studio, and public share links — all funnel through one Vercel proxy into API Gateway. Reads resolve against Aurora DSQL; the meter writes to DynamoDB; audio streams straight from signed CloudFront. The same API is agent-native via x402 + MCP.

Money in, money out, and who's who

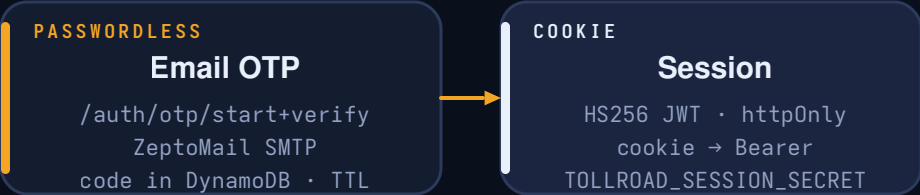
WALLET TOP-UP



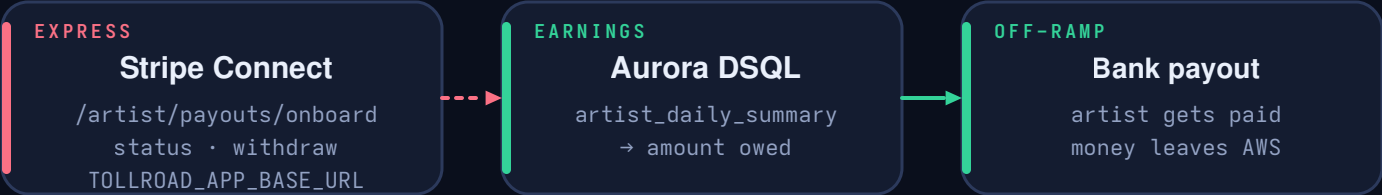
x402 · AGENT-NATIVE PAYMENT



IDENTITY



ARTIST PAYOUTS



Money in: Stripe top-ups land via a key-exempt webhook → a DynamoDB TOPUP event → the projector credits Aurora DSQL.
Identity is passwordless email OTP (ZeptoMail) → an HS256 session cookie. Money out: an artist's DSQL earnings settle to their bank via Stripe Connect.

Every resource, one table

RESOURCE	TYPE	TRIGGER / SOURCE	TALKS TO
TollroadStack	CDK Stack	cdk deploy · us-east-1	every resource
tollroad	DynamoDB	REST + Streams · on-demand	api · projector
GSI1	DDB index	ARTIST#id · reverse	earnings lookups
Stream	DDB stream	NEW_AND_OLD_IMAGES	ProjectorConsumerFn
TollroadDsqli	Aurora DSQL	IAM admin + projector role	api + projector
tollroad-api	API Gateway	REST · stage v1 · API-key	tollroad-api λ
ApiFn (tollroad-api)	Lambda	API GW proxy · ESM	DDB · DSQL · S3 · Bedrock
ProjectorConsumerFn	Lambda	DDB Stream · batch 100	DynamoDB → Aurora DSQL
DsqiDepsLayer	Lambda Layer	pg · @aws-sdk/dsql-signer	projector
JWT authorizer	Lambda	REQUEST authorizer (opt)	API Gateway
TollroadAudioBucket	S3 bucket	SSE-KMS · OAC · block public	CloudFront · presign
TollroadCdn	CloudFront	signed URL · trusted KeyGroup	S3 audio origin
TollroadImagesBucket	S3 bucket	covers · avatars	TollroadImagesCdn
TollroadImagesCdn	CloudFront	public · OAC	S3 images origin
TollroadAudioKey	KMS CMK	alias/tollroad-audio · rotate	S3 · CloudFront OAC
Bedrock	AI	titan-embed-text-v2 · 1024-d	tollroad-api (vibe search)
Stripe	External	PaymentIntent + webhook	wallet · payouts
ZeptoMail	External SMTP	OTP delivery · :587	auth
Secrets	Env / config	session · CF key · SMTP · Stripe	api λ
MCP server	External	agent tool surface	metered API (x402)

polyglot by design · DynamoDB = command/write side · Aurora DSQL = query/read side · CQRS bridged by one projector · single region · one CDK stack

The complete production inventory — construct, type, what fires it, and what it touches.
Two AWS databases chosen on purpose: DynamoDB for the high-write metering ledger, Aurora DSQL for relational truth — joined by exactly one CQRS projector.